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Program Check Report

Disclaimer:
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Report Summary

Overall result: Violation

Program: App_DoughFactory
Source: robot_configuration20261014_19_16_56.zip
Parsed 1,921 script lines, 38 MoveJ, 68 MoveL, 0 MoveP, 57 blended motions, 14 payload calls, and 2 TCP calls.

URCaps Used in This Program

UR Connect, 2.3.8, Universal Robots - node type: UR Connect - script line 25

Compliance Status

In Compliance	Follows Universal Robots recommended programming practices.
Potential Concern	Could potentially violate Universal Robots recommendations depending on the application.
Violation	Does not follow Universal Robots recommendations.

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Checklist Details		
Initial Payload Mass, CoG and Inertia	Initial payload is set to 3.6 kg with CoG 0.001 m, 0 m, 0.132 m and inertia values.	✓
Dynamic payload changes	14 payload calls were found with changing values.	✓
Initial TCP	Initial TCP is set before motion starts to p[0, 0, 0.247, 0, 3.1414, 0].	✓
Dynamic TCP changes	2 TCP calls were found with changing values.	✓
Acceleration limits	One or more motion accelerations are above the recommendation.	✗
Blend radius overlap	One or more blend radii are overlapping or need manual review because the target position is dynamic.	✗
Blended motion continuation	One or more blended motions are followed by commands that can abort blending.	!

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Detailed findings

Each section below keeps the current URacle Program Check content while using the Universal Robots performance-check report style.

Initial Payload Mass, CoG and Inertia

Status: In Compliance

Initial payload is set to 3.6 kg with CoG 0.001 m, 0 m, 0.132 m and inertia values.

Findings

No findings: No issues detected for this recommendation.

Dynamic payload changes

Status: In Compliance

14 payload calls were found with changing values.

Findings

No findings: No issues detected for this recommendation.

Initial TCP

Status: In Compliance

Initial TCP is set before motion starts to p[0, 0, 0.247, 0, 3.1414, 0].

Findings

No findings: No issues detected for this recommendation.

Dynamic TCP changes

Status: In Compliance

2 TCP calls were found with changing values.

Findings

No findings: No issues detected for this recommendation.

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Acceleration limits

Status: Violation

One or more motion accelerations are above the recommendation.

Findings

MoveL acceleration exceeds 2500 mm/s2

Line: Script 448 / PolyScope 313.

MOVEL uses 3000 mm/s2.

Recommended action: Reduce MoveL acceleration to 2500 mm/s2 or less.

Blend radius overlap

Status: Violation

One or more blend radii are overlapping or need manual review because the target position is dynamic.

Findings

Blend radii overlap

Line: Script 598 / PolyScope 380.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 3 mm and 3 mm, for 6 mm combined radius. The radii overlap by 6 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 615 / PolyScope 390.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 80 mm and 80 mm, for 160 mm combined radius. The radii overlap by 160 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 622 / PolyScope 393.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 50 mm and 50 mm, for 100 mm combined radius. The radii overlap by 100 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 636 / PolyScope 399.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 80 mm and 80 mm, for 160

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Blend radius overlap (continued)

mm combined radius. The radii overlap by 160 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 643 / PolyScope 403.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 50 mm and 50 mm, for 100 mm combined radius. The radii overlap by 100 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 736 / PolyScope 448.

MOVEL to place_above and MOVEL to place_above are 0 mm apart. Their blend radii are 50 mm and 50 mm, for 100 mm combined radius. The radii overlap by 100 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 737 / PolyScope 448.

MOVEL to place_above and MOVEL to place_above are 0 mm apart. Their blend radii are 50 mm and 50 mm, for 100 mm combined radius. The radii overlap by 100 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 740 / PolyScope 450.

MOVEL to place_above and MOVEL to place_above are 0 mm apart. Their blend radii are 50 mm and 50 mm, for 100 mm combined radius. The radii overlap by 100 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 806 / PolyScope 473.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 3 mm and 3 mm, for 5 mm combined radius. The radii overlap by 5 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

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Blend radius overlap (continued)

Line: Script 827 / PolyScope 487.

MOVEL and MOVEL are 0 mm apart. Their blend radii are 3 mm and 3 mm, for 5 mm combined radius. The radii overlap by 5 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 842 / PolyScope 497.

MOVEL to pre_idle and MOVEL to pre_idle are 0 mm apart. Their blend radii are 10 mm and 10 mm, for 20 mm combined radius. The radii overlap by 20 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii overlap

Line: Script 1792 / PolyScope 109.

MOVEL to pre_idle and MOVEL to pre_idle are 0 mm apart. Their blend radii are 10 mm and 10 mm, for 20 mm combined radius. The radii overlap by 20 mm.

Recommended action: Reduce the blend radius until there is no overlap, or remove the blend entirely.

Blend radii need manual review

Line: Script 137.

35 blended motions could not be checked because the target position was computed dynamically.

Recommended action: Review dynamically computed blended targets manually.

Blended motion continuation

Status: Potential Concern

One or more blended motions are followed by commands that can abort blending.

Findings

Blended motion is followed by a motion-aborting command

Line: Script 138 / PolyScope 189.

MOVEJ has a blend radius, but the next executable command is "if (loading_totes()):" at script line 141.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

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Blended motion continuation (continued)

Line: Script 329 / PolyScope 272.

MOVEJ has a blend radius, but the next executable command is "while (not(is_steady())):" at script line 331.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 599 / PolyScope 380.

MOVEL has a blend radius, but the next executable command is "tote_count = tote_count + 1" at script line 606.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 618 / PolyScope 391.

MOVEJ has a blend radius, but the next executable command is "if (stack_side == STACK_SIDE.LEFT):" at script line 621.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 623 / PolyScope 393.

MOVEL has a blend radius, but the next executable command is "right_return = False" at script line 625.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 637 / PolyScope 399.

MOVEL has a blend radius, but the next executable command is "optimovej(transition_B_R_q, v=1, a=1, r=0.15)" at script line 639.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 644 / PolyScope 403.

MOVEL has a blend radius, but the next executable command is "right_return = False" at script line 646.

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Blended motion continuation (continued)

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 657 / PolyScope 409.

MOVEJ has a blend radius, but the next executable command is "if (stack_side == STACK_SIDE.LEFT):" at script line 660.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 664 / PolyScope 412.

MOVEJ has a blend radius, but the next executable command is "if (stack_side == STACK_SIDE.RIGHT):" at script line 670.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 672 / PolyScope 415.

MOVEJ has a blend radius, but the next executable command is "if (stack_side == STACK_SIDE.LEFT):" at script line 675.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 679 / PolyScope 418.

MOVEJ has a blend radius, but the next executable command is "if (loading_trays()):" at script line 688.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 807 / PolyScope 473.

MOVEL has a blend radius, but the next executable command is "if (loading_totes()):" at script line 812.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

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Blended motion continuation (continued)

Blended motion is followed by a motion-aborting command

Line: Script 831 / PolyScope 489.

MOVEJ has a blend radius, but the next executable command is "robot_placing = False" at script line 836.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 843 / PolyScope 497.

MOVEL to pre_idle has a blend radius, but the next executable command is "started_with_gripper_loaded = False" at script line 845.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1758 / PolyScope 91.

MOVEL has a blend radius, but the next executable command is "pull_out_pose[X] = -0.05" at script line 1763.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1765 / PolyScope 96.

MOVEL has a blend radius, but the next executable command is "if (tool_in_conveyor_area() and tool_flat()):" at script line 1768.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1782 / PolyScope 104.

MOVEL has a blend radius, but the next executable command is "current_pose[Z] = 0.1" at script line 1785.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1787 / PolyScope 106.

MOVEL has a blend radius, but the next executable command is "current_pose[Y]

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Blended motion continuation (continued)

= -0.6" at script line 1789.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1793 / PolyScope 109.

MOVEL to pre_idle has a blend radius, but the next executable command is "if (get_actual_tcp_pose ()[0] > -0.57 and get_actual_tcp_pose ()[1] > 0.2):" at script line 1799.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Blended motion is followed by a motion-aborting command

Line: Script 1803 / PolyScope 113.

MOVEJ has a blend radius, but the next executable command is "optimovej(pre_idle_q, v=0.5, a=0.5)" at script line 1806.

Recommended action: Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Recommended Action Summary

1. Reduce MoveL acceleration to 2500 mm/s2 or less.

Applies to Acceleration limits. Examples: script 448 / PolyScope 313

2. Reduce the blend radius until there is no overlap, or remove the blend entirely.

Applies to Blend radius overlap. Examples: script 598 / PolyScope 380; script 615 / PolyScope 390; script 622 / PolyScope 393...

3. Review dynamically computed blended targets manually.

Applies to Blend radius overlap. Examples: script 137

4. Review whether this command should interrupt blending. If smooth continuation is expected, follow the blended move directly with another motion command.

Applies to Blended motion continuation. Examples: script 138 / PolyScope 189; script 329 / PolyScope 272; script 599 / PolyScope 380...

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